

Arc length examples — 2.4

Recall: the arc length from $(a, f(a))$ to $(b, f(b))$

is given by $\int_a^b \sqrt{1 + f'(x)^2} dx$.

Example: Calculate the arc length of the graph of $f(x) = 2x^{3/2}$ over $[0, 1]$.

$$\rightarrow \int_0^1 \sqrt{1 + 9x} dx$$

$$u = 1 + 9x$$

$$du = 9 dx$$

$$\begin{cases} f'(x) = 3x^{1/2} \\ f'(x)^2 = 9x \end{cases}$$

$$= \frac{1}{9} \int_1^{10} \sqrt{u} du = \frac{1}{9} \frac{u^{3/2}}{3/2} \Big|_1^{10} = \frac{2}{27} u^{3/2} \Big|_1^{10} = \frac{2}{27} (10\sqrt{10} - 1)$$

Example: Calculate the arc length of the graph of $f(x) = x^2$ over $[0, 4]$.

$$f'(x) = 2x, \quad f'(x)^2 = 4x^2$$

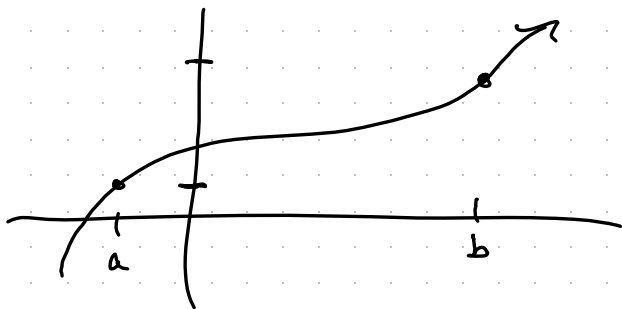
$$\text{length} = \int_0^4 \sqrt{1 + 4x^2} \, dx$$

$$x = \frac{1}{2} \tan \theta$$

$$dx = \frac{1}{2} \sec^2 \theta \, d\theta$$

$$= \int_0^{\arctan(8)} \sqrt{1 + \tan^2 \theta} \cdot \frac{1}{2} \sec^2 \theta \, d\theta = \int_0^{\arctan(8)} \frac{1}{2} \sec^3 \theta \, d\theta = \dots$$

$$= 2\sqrt{65} + \frac{1}{2} \ln |\sqrt{65} + 8|$$



Wanting length of
 $f(x) = x^2$ over $[0, 4]$
 is the same as
 wanting length of
 $g(y) = \sqrt{y}$ over $[0, 16]$

$$\text{Arc length} = \int_0^{16} \sqrt{1 + \frac{1}{4y}} dy.$$